Numerical Physics — Exercise 5

Equation d'onde dans un milieu inhomogène : modes propres et propagation d'une vague de tsunami

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Contents

1 Introduction

This report studies the numerical solution of one-dimensional wave equations in media with constant and spatially varying propagation speed. We first implement and validate explicit finite-difference schemes for three formulations of the wave equation. We then analyse reflection, eigenmodes, resonance and stability in the constant-speed case, before applying the method to tsunami propagation over an inhomogeneous bathymetry.
In the last part of the report, we focus on the tsunami configuration. In this case, the propagation speed is not prescribed directly, but obtained from the local depth through

$$u(x) = \sqrt{gh(x)}.$$

This makes the three wave equations behave differently, even though they coincide in the constant-speed case. The numerical results are compared with WKB predictions in order to understand how the amplitude changes when the wave enters shallow water and how the validity of this approximation depends on the smoothness of the bathymetry.

2 Numerical method and implementation

The aim of the numerical implementation is to solve the one-dimensional wave equation in a medium with spatially dependent propagation velocity $u(x)$. Three different models are considered:

$$\text{Equation A:} \qquad \frac{\partial^2 f}{\partial t^2} = u^2(x)\frac{\partial^2 f}{\partial x^2}, \tag{1}$$

$$\text{Equation B:} \qquad \frac{\partial^2 f}{\partial t^2} = \frac{\partial}{\partial x}\left(u^2(x)\frac{\partial f}{\partial x}\right), \tag{2}$$

$$\text{Equation C:} \qquad \frac{\partial^2 f}{\partial t^2} = \frac{\partial^2}{\partial x^2}\left(u^2(x)f\right). \tag{3}$$

Here, $f(x,t)$ represents the wave perturbation and $u(x)$ is the local propagation velocity. In the tsunami application, this velocity is related to the local water depth through the shallow-water approximation,

$$u(x) = \sqrt{gh(x)}, \tag{4}$$

where g is the gravitational acceleration and $h(x)$ is the water depth.

2.1 Spatial and temporal discretization

The spatial domain $x \in [0, L]$ is discretized using a uniform grid of $N = n_x + 1$ points,

$$x_i = i\Delta x, \qquad \Delta x = \frac{L}{n_x}, \qquad i = 0, \ldots, n_x.$$

The time domain is discretized with a constant time step Δt, so that $t^n = n\Delta t$. We denote the numerical approximation of $f(x_i, t^n)$ by f_i^n.
The second-order time derivative is approximated using the standard explicit three-level centered scheme,

$$\frac{\partial^2 f}{\partial t^2}(x_i, t^n) \approx \frac{f_i^{n+1} - 2f_i^n + f_i^{n-1}}{\Delta t^2}.$$

This gives an explicit update formula of the form

$$f_i^{n+1} = 2f_i^n - f_i^{n-1} + \Delta t^2\, \mathrm{RHS}_i^n, \tag{5}$$

where RHS_i^n depends on the equation considered.
The spatial derivatives are approximated using centered finite differences,

$$\frac{\partial g}{\partial x}(x_i) \approx \frac{g_{i+1} - g_{i-1}}{2\Delta x}, \qquad \frac{\partial^2 g}{\partial x^2}(x_i) \approx \frac{g_{i+1} - 2g_i + g_{i-1}}{\Delta x^2}.$$

These approximations are second-order accurate in space. Since the time derivative is also discretized with a centered second-order formula, the full scheme is formally second-order accurate in both space and time, provided that the solution is sufficiently smooth.

2.2 Discrete schemes for the three equations

For Equation A, the right-hand side is simply $u^2 f_{xx}$. Defining

$$\beta_i^2 = \frac{u_i^2 \Delta t^2}{\Delta x^2},$$

the update formula becomes

$$f_i^{n+1} = 2f_i^n - f_i^{n-1} + \beta_i^2 \left(f_{i+1}^n - 2f_i^n + f_{i-1}^n \right).$$

For Equation B, the right-hand side is

$$\frac{\partial}{\partial x} \left(u^2 \frac{\partial f}{\partial x} \right).$$

In the implementation, this term is written using the product rule as

$$\frac{\partial}{\partial x} \left(u^2 \frac{\partial f}{\partial x} \right) = u^2 \frac{\partial^2 f}{\partial x^2} + \frac{\partial u^2}{\partial x} \frac{\partial f}{\partial x}.$$

Both derivatives are then discretized using centered finite differences,

$$\left(\frac{\partial u^2}{\partial x} \right)_i \approx \frac{u_{i+1}^2 - u_{i-1}^2}{2\Delta x}, \qquad \left(\frac{\partial f}{\partial x} \right)_i \approx \frac{f_{i+1}^n - f_{i-1}^n}{2\Delta x},$$

and

$$\left(\frac{\partial^2 f}{\partial x^2} \right)_i \approx \frac{f_{i+1}^n - 2f_i^n + f_{i-1}^n}{\Delta x^2}.$$

Therefore, the numerical update used for Equation B is

$$f_i^{n+1} = 2f_i^n - f_i^{n-1} + \Delta t^2 \left[u_i^2 \frac{f_{i+1}^n - 2f_i^n + f_{i-1}^n}{\Delta x^2} + \frac{u_{i+1}^2 - u_{i-1}^2}{2\Delta x} \frac{f_{i+1}^n - f_{i-1}^n}{2\Delta x} \right].$$

For Equation C, the right-hand side is the second derivative of the product $u^2 f$. We therefore discretize it as

$$\frac{\partial^2}{\partial x^2} \left(u^2 f \right)_i \approx \frac{u_{i+1}^2 f_{i+1}^n - 2u_i^2 f_i^n + u_{i-1}^2 f_{i-1}^n}{\Delta x^2}.$$

The corresponding update formula is

$$f_i^{n+1} = 2f_i^n - f_i^{n-1} + \frac{\Delta t^2}{\Delta x^2} \left(u_{i+1}^2 f_{i+1}^n - 2u_i^2 f_i^n + u_{i-1}^2 f_{i-1}^n \right).$$

The three schemes are identical when $u(x)$ is constant. In that case, Equations A, B and C all reduce to the standard one-dimensional wave equation,

$$\frac{\partial^2 f}{\partial t^2} = u^2 \frac{\partial^2 f}{\partial x^2}.$$

2.3 Implementation details

The numerical implementation is organized around three vectors f^{n-1}, f^n and f^{n+1}, stored in the code as `fpast`, `fnow` and `fnext`. At each time step, the interior points are first updated using the selected equation type, A, B or C. Then the boundary conditions are imposed. Finally, the time levels are shifted according to

$$f^{n-1} \leftarrow f^n, \qquad f^n \leftarrow f^{n+1}.$$

The code also stores the grid points x_i, the water depth profile $h(x_i)$, the squared velocity $u_i^2 = gh_i$, and the local squared CFL factor,

$$\beta_i^2 = \frac{u_i^2 \Delta t^2}{\Delta x^2}.$$

The output files contain the spatial grid, the velocity profile, the wave profile at selected time steps, and the wave energy as a function of time. To avoid generating excessively large files, the wave field $f(x,t)$ is written only every `n_stride` time steps.

3 Simulations and analyses

3.1 Constant propagation speed

3.1.1 a) Reflection at the boundaries

We consider the constant propagation speed case,

$$L = 20\,\mathrm{m}, \qquad u = 6\,\mathrm{m\,s^{-1}}. \tag{6}$$

Since $u(x)$ is constant, equations A, B and C become identical. Therefore, in this section we use equation A for the simulations. The left boundary is harmonically driven:

$$f(0,t) = A\sin(\omega t), \qquad \omega = 5\,\mathrm{rad\,s^{-1}}. \tag{7}$$

The transit time of the wave through the domain is

$$t_{\mathrm{tr}} = \frac{L}{u} = \frac{20}{6} \simeq 3.33\,\mathrm{s}. \tag{8}$$

Thus, two round trips correspond to

$$4t_{\mathrm{tr}} \simeq 13.33\,\mathrm{s}. \tag{9}$$

We therefore used $t_{\mathrm{fin}} = 15\,\mathrm{s}$, which is long enough to observe two round trips of the wave.
To illustrate the reflection phenomenon at the right boundary, we first compare fixed and free boundary conditions. For a fixed boundary,

$$f(L,t) = 0, \tag{10}$$

the reflected wave is expected to be phase-inverted. For a free boundary,

$$\frac{\partial f}{\partial x}(L,t) = 0, \tag{11}$$

the reflected wave keeps the same sign. This difference is shown in Figure 1, where several profiles $f(x,t)$ are plotted around and after the first impact with the right boundary.

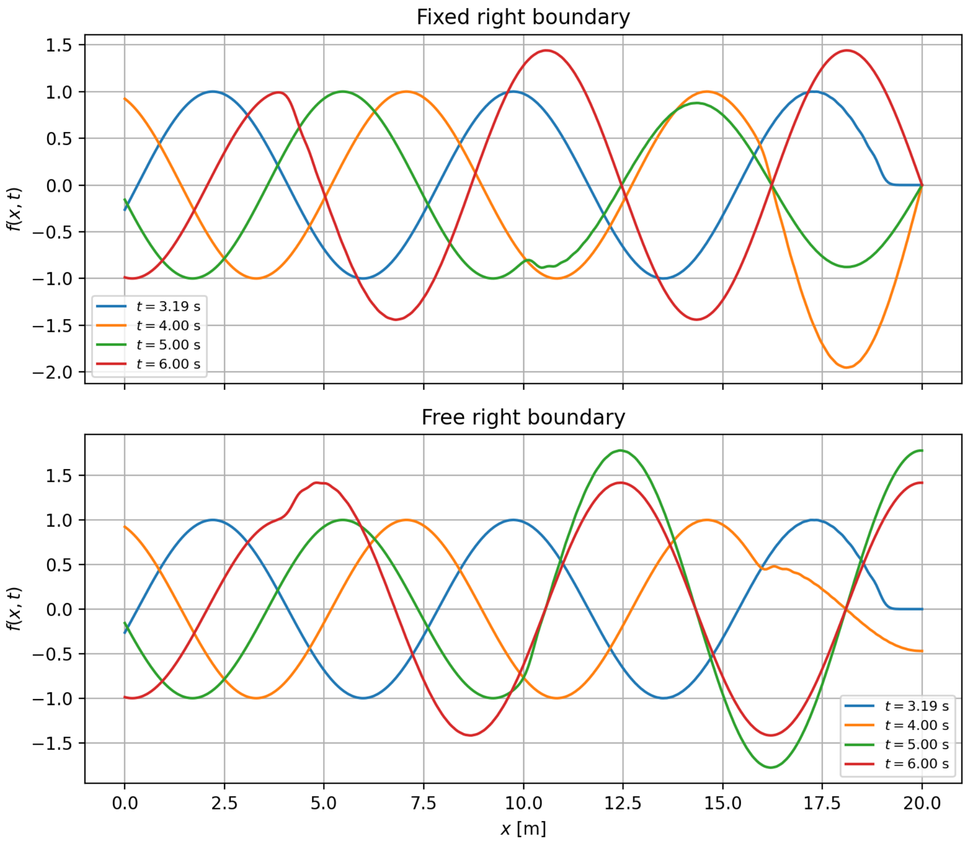


Figure 1: Comparison between fixed and free right boundary conditions. The profiles are plotted around the first reflection time $t_{\mathrm{tr}} \simeq 3.33\,\mathrm{s}$. For the fixed boundary, the reflected wave is inverted in sign, while for the free boundary the reflected wave keeps the same sign. Simulation parameters: $n_x = 400$, $\beta_{\mathrm{CFL}} = 0.9$.

We then consider the outgoing-wave boundary condition at the right boundary. This condition is designed to let the wave leave the computational domain while reducing artificial reflections. Figure 2 shows the corresponding space-time diagram. In contrast with the fixed and free boundary conditions, the reflected signal is strongly reduced.

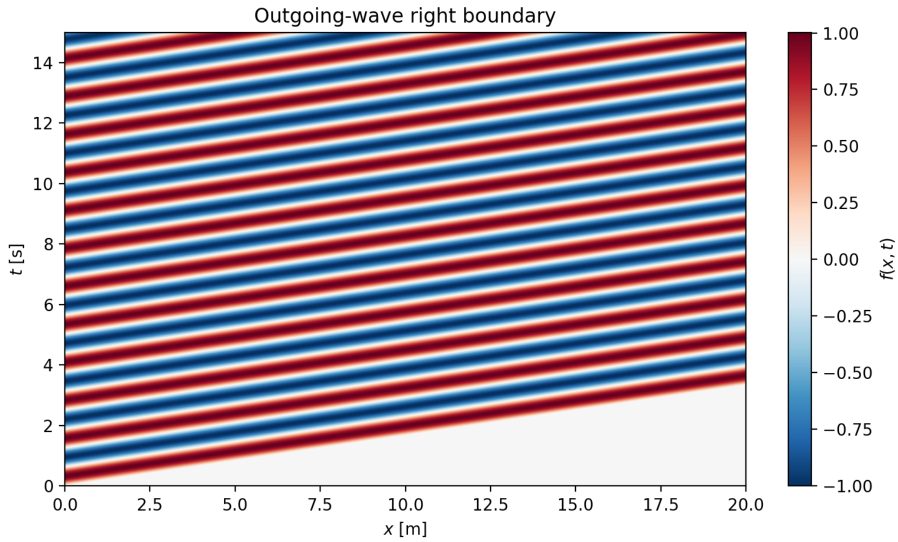


Figure 2: Space-time diagram for the outgoing-wave right boundary condition. The wave leaves the domain at $x = L$, and artificial reflections are strongly reduced. Simulation parameters: $n_x = 400$, $\beta_{\rm CFL} = 0.9$, $t_{\rm fin} = 15\,$s.

Finally, we performed a convergence study for the outgoing-wave boundary condition. The solution was evaluated at

$$x = 5\,\text{m}, \qquad t = 1.5\,\text{s}. \tag{12}$$

The spatial resolution was varied, while keeping the CFL number fixed. Since

$$\Delta t = \beta_{\rm CFL} \frac{\Delta x}{u}, \tag{13}$$

refining the spatial grid also refines the time step.
At this point in space and time, the signal has not yet been affected by a reflection from the right boundary. Indeed, the wave reaches the right boundary at $t = L/u \simeq 3.33\,$s, whereas here $t = 1.5\,$s. Therefore, the numerical result can be compared with the travelling-wave solution

$$f_{\rm exact}(x,t) = \sin\left[\omega\left(t - \frac{x}{u}\right)\right]. \tag{14}$$

The absolute error was computed as

$$\varepsilon(\Delta x) = \left| f_{\Delta x}(5\,\text{m}, 1.5\,\text{s}) - f_{\rm exact}(5\,\text{m}, 1.5\,\text{s}) \right|. \tag{15}$$

n_x	Δx	Δt	$f(5\,\text{m}, 1.5\,\text{s})$	Error
50	$4.000 \cdot 10^{-1}$	$6.000 \cdot 10^{-2}$	$-1.872 \cdot 10^{-1}$	$3.365 \cdot 10^{-3}$
100	$2.000 \cdot 10^{-1}$	$3.000 \cdot 10^{-2}$	$-1.901 \cdot 10^{-1}$	$4.532 \cdot 10^{-4}$
200	$1.000 \cdot 10^{-1}$	$1.500 \cdot 10^{-2}$	$-1.893 \cdot 10^{-1}$	$1.265 \cdot 10^{-3}$
400	$5.000 \cdot 10^{-2}$	$7.500 \cdot 10^{-3}$	$-1.903 \cdot 10^{-1}$	$2.357 \cdot 10^{-4}$
800	$2.500 \cdot 10^{-2}$	$3.750 \cdot 10^{-3}$	$-1.906 \cdot 10^{-1}$	$3.748 \cdot 10^{-6}$
1600	$1.250 \cdot 10^{-2}$	$1.875 \cdot 10^{-3}$	$-1.905 \cdot 10^{-1}$	$3.674 \cdot 10^{-5}$

Table 1: Convergence study for the outgoing-wave boundary condition at $x = 5\,$m, $t = 1.5\,$s, with fixed $\beta_{\rm CFL} = 0.9$. The error is computed with respect to the analytical travelling-wave solution.

The convergence curve is shown in Figure 3. The error shows an overall decreasing trend as the grid is refined, although the decrease is not perfectly monotonic for all resolutions. This is likely due to the interpolation used to evaluate the solution at exactly $x = 5\,\mathrm{m}$, $t = 1.5\,\mathrm{s}$, and to small phase errors in the numerical travelling wave.

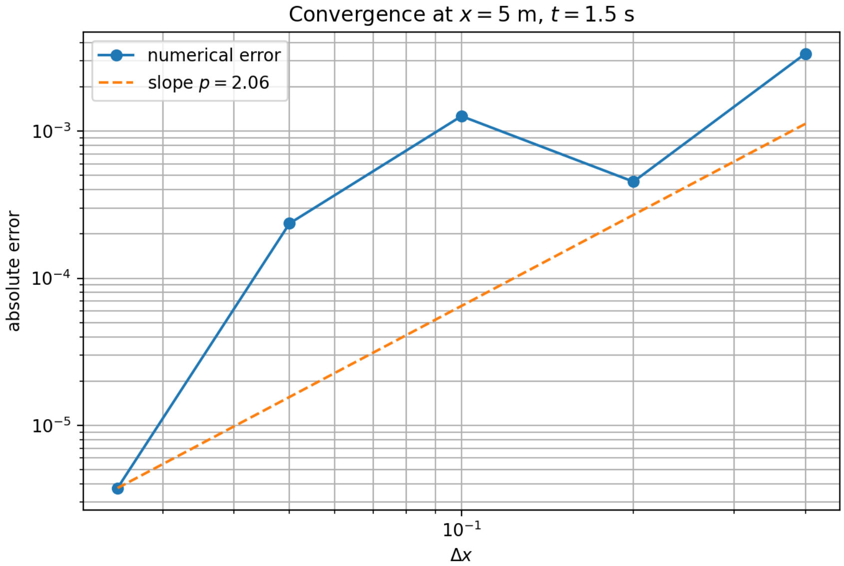


Figure 3: Convergence of $f(5\,\mathrm{m}, 1.5\,\mathrm{s})$ for the outgoing-wave boundary condition. The error is computed with respect to the analytical travelling-wave solution.

Using a log-log fit of the error as a function of Δx, the estimated convergence order is

$$p \simeq 2.06. \tag{16}$$

Despite the small non-monotonic variations in the error, this estimate is compatible with the expected second-order accuracy of the finite-difference scheme. The simulations reproduce the expected qualitative behaviour of the three boundary conditions: a fixed boundary reflects the wave with a sign inversion, a free boundary reflects it without sign inversion, and the outgoing-wave condition strongly reduces artificial reflections. The convergence study also shows that the numerical solution stabilizes as Δx and Δt are decreased at fixed β_{CFL}.

3.1.2 b) Eigenmodes

We now compute analytically the eigenfrequencies and eigenmodes of the system without external excitation. In the simulations, the left boundary is harmonically driven and the right boundary is fixed. However, for the computation of the natural modes of the system, the excitation is removed. Therefore, both boundaries are taken to be fixed:

$$f(0,t) = 0, \qquad f(L,t) = 0. \tag{17}$$

Since the propagation speed is constant, the three equations A, B and C are identical and reduce to the homogeneous wave equation

$$\frac{\partial^2 f}{\partial t^2} = u^2 \frac{\partial^2 f}{\partial x^2}. \tag{18}$$

We look for separated solutions of the form

$$f(x,t) = X(x)T(t). \tag{19}$$

Substituting this expression into Eq. (18) gives

$$X(x)T''(t) = u^2 X''(x)T(t).$$

Dividing by $u^2X(x)T(t)$, we obtain

$$\frac{T''(t)}{u^2T(t)} = \frac{X''(x)}{X(x)} = -k^2. \tag{20}$$

This leads to the two ordinary differential equations

$$X''(x) + k^2X(x) = 0, \tag{21}$$

and

$$T''(t) + u^2k^2T(t) = 0. \tag{22}$$

The spatial solution is

$$X(x) = C_1 \sin(kx) + C_2 \cos(kx).$$

The fixed boundary condition at $x = 0$ gives

$$X(0) = 0,$$

so

$$C_2 = 0.$$

The fixed boundary condition at $x = L$ gives

$$X(L) = 0,$$

therefore

$$C_1 \sin(kL) = 0.$$

For non-trivial solutions, $C_1 \neq 0$, so

$$\sin(kL) = 0.$$

Thus, the allowed wavenumbers are

$$k_n = \frac{n\pi}{L}, \qquad n = 1, 2, 3, \ldots \tag{23}$$

The corresponding spatial eigenmodes are

$$X_n(x) = \sin\left(\frac{n\pi x}{L}\right). \tag{24}$$

The temporal equation can be written as

$$T_n''(t) + \omega_n^2T_n(t) = 0,$$

with

$$\omega_n = uk_n.$$

Using Eq. (23), the angular eigenfrequencies are

$$\boxed{\omega_n = \frac{n\pi u}{L}}, \qquad n = 1, 2, 3, \ldots \tag{25}$$

Therefore, the eigenmodes of the full problem can be written as

$$\boxed{f_n(x,t) = \sin\left(\frac{n\pi x}{L}\right)[A_n \cos(\omega_n t) + B_n \sin(\omega_n t)]}. \tag{26}$$

In the present case,

$$L = 20\,\mathrm{m}, \qquad u = 6\,\mathrm{m\,s^{-1}}.$$

Therefore,

$$\omega_n = \frac{n\pi \cdot 6}{20} = \frac{3\pi}{10} n \simeq 0.94248\, n \ \mathrm{rad\,s^{-1}}. \tag{27}$$

The first eigenfrequencies are therefore

n	ω_n [rad s^{-1}]
1	0.942
2	1.885
3	2.827
4	3.770
5	4.712
6	5.655
7	6.597
8	7.540

Equivalently, the ordinary frequencies in Hz are

$$\nu_n = \frac{\omega_n}{2\pi} = \frac{nu}{2L}. \tag{28}$$

With $u = 6\,\mathrm{m\,s^{-1}}$ and $L = 20\,\mathrm{m}$, this gives

$$\nu_n = 0.15n \ \mathrm{Hz}.$$

These eigenmodes will be used in the next section to interpret the resonant response of the system when the harmonic forcing frequency is close to one of the natural frequencies ω_n.

3.1.3 c) Excitation of eigenmodes

We now study how the system responds to a harmonic excitation at the left boundary,

$$f(0,t) = A\sin(\omega t), \tag{29}$$

with a fixed boundary condition at the right boundary,

$$f(L,t) = 0. \tag{30}$$

The propagation speed is constant, $u = 6\,\mathrm{m\,s^{-1}}$, and the domain length is $L = 20\,\mathrm{m}$. Therefore, the analytical eigenfrequencies are

$$\omega_n = \frac{n\pi u}{L} = \frac{3\pi}{10} n, \qquad n = 1, 2, 3, \ldots \tag{31}$$

as derived in the previous section.

For each forcing frequency ω, we compute the maximum value of the energy over the simulation time:

$$\hat{E}(\omega) = \max_{t\in[0,t_{\mathrm{fin}}]} E(t)\big|_{\omega}. \tag{32}$$

This quantity measures the amplitude of the resonant response of the system.

The frequency range was chosen to include several analytical eigenfrequencies. The vertical dashed lines in Figure 4 indicate the theoretical values of ω_n. We observe clear peaks in $\hat{E}(\omega)$ when the forcing frequency is close to one of the eigenfrequencies. This is the expected resonance phenomenon: when $\omega \simeq \omega_n$, the harmonic excitation efficiently transfers energy into the corresponding normal mode.

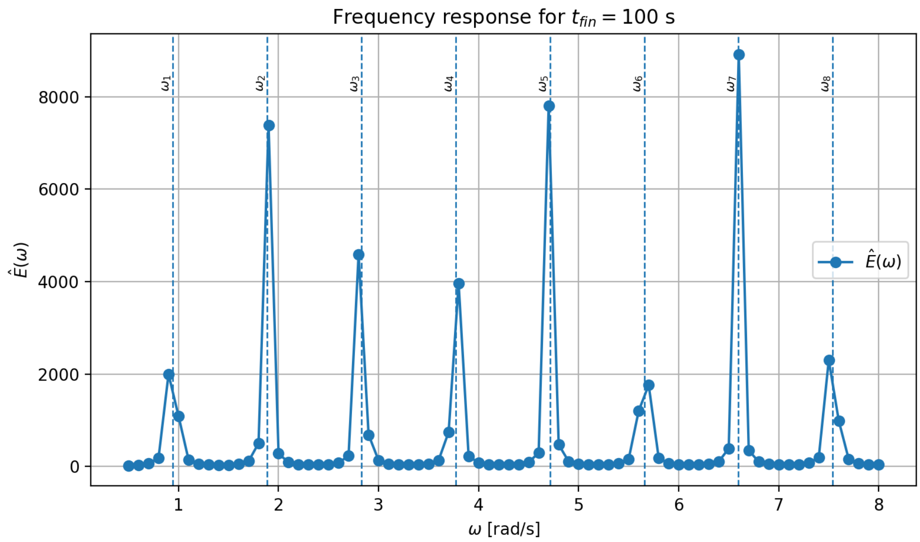


Figure 4: Maximum energy $\hat{E}(\omega)$ as a function of the forcing frequency. Peaks appear close to the analytical eigenfrequencies $\omega_n = n\pi u/L$. Simulation parameters: $n_x = 400$, $\beta_{\text{CFL}} = 0.9$, $t_{\text{fin}} = 100\,\text{s}$.

To study the effect of the simulation duration, we repeated the frequency scan for several values of t_{fin}. The transit time is

$$t_{\text{tr}} = \frac{L}{u} \simeq 3.33\,\text{s}. \tag{33}$$

Thus, the chosen final times correspond to several tens of transits. The results are shown in Figure 5. As t_{fin} increases, the peaks become more pronounced and narrower. This is consistent with the fact that near resonance, the system has more time to accumulate energy.

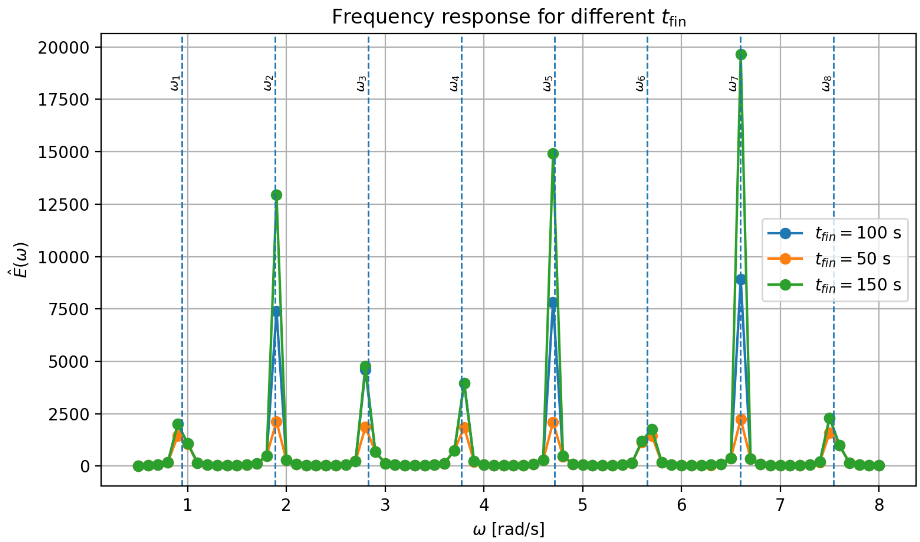


Figure 5: Comparison of the frequency response $\hat{E}(\omega)$ for different final times t_{fin}. Longer simulations lead to sharper and stronger resonant peaks. Simulation parameters: $n_x = 400$, $\beta_{\text{CFL}} = 0.9$.

For the frequency range and resolution used in this scan, the largest response was obtained close to

$$\omega \simeq 6.60\,\text{rad}\,\text{s}^{-1}. \tag{34}$$

This value is close to the analytical eigenfrequency

$$\omega_7 = 6.597\,\mathrm{rad\,s^{-1}}. \tag{35}$$

Therefore, the dominant excited mode is expected to be approximately the mode $n = 7$.
Finally, we compare the numerical solution at a time close to t_{fin} with the corresponding analytical eigenmode,

$$X_n(x) = \sin\left(\frac{n\pi x}{L}\right). \tag{36}$$

Since the numerical solution may have an arbitrary amplitude and sign, both the numerical profile and the analytical mode are normalized before comparison:

$$\tilde{f}(x,t) = \frac{f(x,t)}{\max_x |f(x,t)|}, \qquad \tilde{X}_n(x) = \frac{X_n(x)}{\max_x |X_n(x)|}. \tag{37}$$

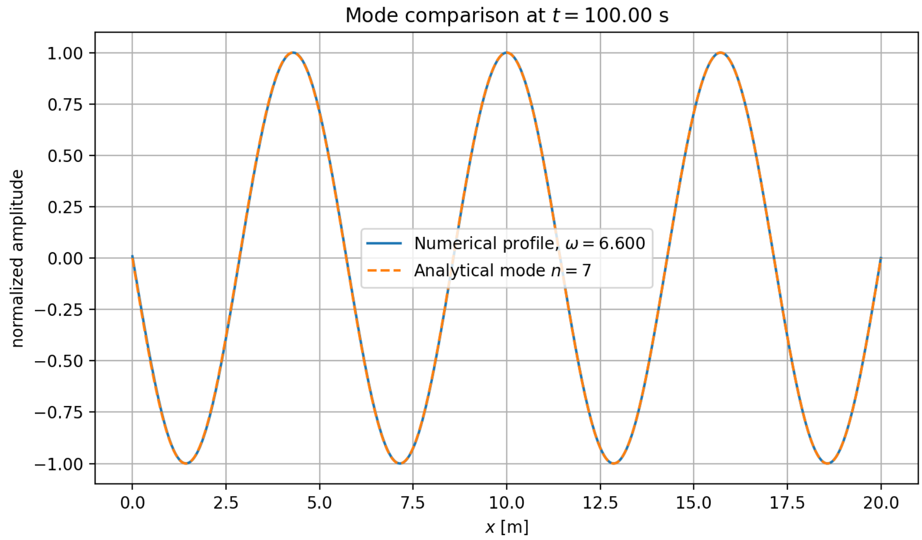


Figure 6: Comparison between the numerical profile $f(x,t)$ near $t = t_{\mathrm{fin}}$ and the normalized analytical eigenmode $X_n(x) = \sin(n\pi x/L)$. The comparison is performed for the dominant resonant mode $n = 7$, at $\omega = 6.60\,\mathrm{rad\,s^{-1}}$, using the profile at $t = 99.9975\,\mathrm{s}$.

The numerical profile agrees well with the analytical eigenmode, confirming that when the forcing frequency is close to an eigenfrequency, the system responds mainly by exciting the corresponding normal mode.

3.1.4 d) Stability

We finally study the effect of the CFL parameter β_{CFL} on the stability of the explicit finite-difference scheme. In the constant-speed case, the time step is chosen as

$$\Delta t = \beta_{\mathrm{CFL}} \frac{\Delta x}{u}. \tag{38}$$

Therefore, increasing β_{CFL} increases the time step while keeping the spatial resolution fixed.
For the one-dimensional wave equation, the explicit three-level scheme is expected to be stable only if the numerical domain of dependence contains the physical domain of dependence. This leads to the CFL stability condition

$$\beta_{\mathrm{CFL}} \le 1. \tag{39}$$

To verify this numerically, we performed simulations with several values of β_{CFL}, keeping the other parameters fixed. We monitored the maximum absolute amplitude of the solution,

$$M(t) = \max_{x} |f(x,t)|. \tag{40}$$

If the scheme is stable, $M(t)$ remains bounded. If the scheme is unstable, small numerical errors grow rapidly and $M(t)$ diverges.

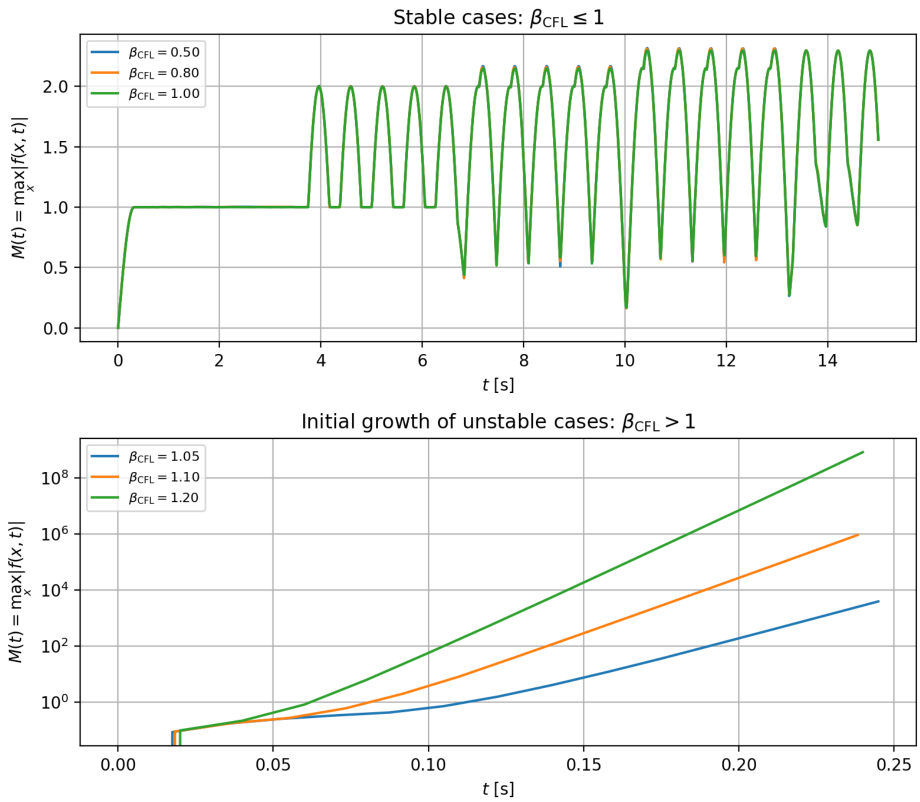


Figure 7: Evolution of $M(t) = \max_x |f(x,t)|$ for different values of β_{CFL}. The upper panel shows the stable cases $\beta_{\text{CFL}} \leq 1$, for which the amplitude remains bounded. The lower panel shows the initial growth for unstable cases $\beta_{\text{CFL}} > 1$, where the amplitude increases rapidly on a logarithmic scale. Simulation parameters: $n_x = 400$, $t_{\text{fin}} = 15\,\text{s}$.

The final amplitudes obtained in the simulations are summarized in Table 2.

β_{CFL}	Δt	Behaviour
0.50	$4.1667 \cdot 10^{-3}$	stable
0.80	$6.6667 \cdot 10^{-3}$	stable
1.00	$8.3333 \cdot 10^{-3}$	stable
1.05	$8.7500 \cdot 10^{-3}$	unstable
1.10	$9.1667 \cdot 10^{-3}$	unstable
1.20	$1.0000 \cdot 10^{-2}$	unstable

Table 2: Effect of β_{CFL} on the stability of the explicit scheme. The stability threshold is observed around $\beta_{\text{CFL}} = 1$.

The results confirm the expected CFL stability criterion. For $\beta_{\text{CFL}} \leq 1$, the numerical solution remains bounded during the simulation. For $\beta_{\text{CFL}} > 1$, the solution becomes unstable and the amplitude grows very rapidly. This behaviour is consistent with the theoretical CFL condition for the explicit scheme.

3.2 Application to a tsunami wave on the Great Barrier Reef

We now apply the numerical schemes to the propagation of a tsunami wave over a non-uniform bathymetry. In the shallow-water approximation, the propagation speed is related to the water depth by

$$u(x) = \sqrt{gh(x)}. \tag{41}$$

Therefore, the wave slows down when it reaches shallow water. This situation is particularly interesting because the three equations A, B and C are no longer equivalent when $u(x)$ varies with space.
The bathymetry used in the simulations is shown in Figure 8. The depth is large far from the reef and becomes much smaller in the central region. This produces a strong decrease of the propagation velocity near the reef.

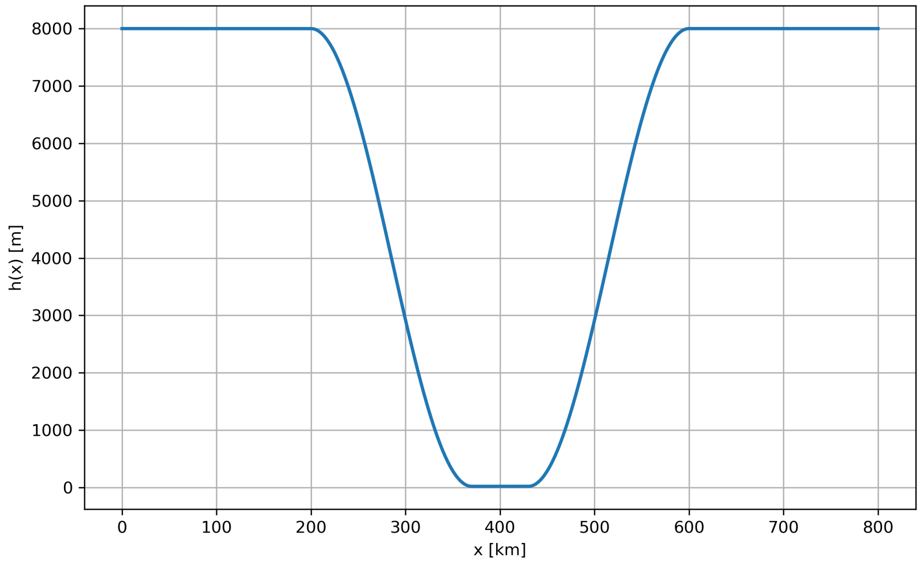


Figure 8: Water depth profile $h(x)$ used for the tsunami simulations. The propagation speed is given by $u(x) = \sqrt{gh(x)}$, so the wave velocity decreases strongly in the shallow region.

The wave is generated at the left boundary using a harmonic excitation,

$$f(0,t) = A\sin(\omega t), \tag{42}$$

with

$$A = 1\,\mathrm{m}, \qquad T = 15\,\mathrm{min} = 900\,\mathrm{s}, \qquad \omega = \frac{2\pi}{T}.$$

At the right boundary we use an outgoing-wave condition in order to let the wave leave the domain with minimal artificial reflection.

3.2.1 a) Comparison between equations A, B and C

We first compare the three possible wave equations in the variable-depth case. In the constant-speed case they all reduce to the same equation, but this is no longer true when $u(x)$ depends on x. The WKB analysis derived in the supplementary material predicts the following amplitude scalings:

$$a_A(x) \propto h(x)^{1/4}, \qquad a_B(x) \propto h(x)^{-1/4}, \qquad a_C(x) \propto h(x)^{-3/4}. \tag{43}$$

These relations imply very different physical behaviours. For equation A, the amplitude is expected to decrease when the wave enters shallow water, because $h^{1/4}$ becomes smaller. For equations B and C, the amplitude is expected to increase in shallow water. The amplification is moderate for equation B and much stronger for equation C.
The numerical simulations confirm this qualitative behaviour. In equation A, the numerical first crest decreases as the wave enters the shallow region. In equation B, the crest amplitude increases moderately, which is consistent with the classical shoaling behaviour predicted by the shallow-water approximation. In equation C, the amplification is much stronger, in agreement with the theoretical scaling proportional to $h^{-3/4}$.
In order to compare the numerical simulations with the WKB prediction, we extracted the first crest amplitude as a function of position. This post-processing is delicate because the wave slows down considerably in the shallow region and the crest becomes harder to track after the interaction with the reef. For this reason, the comparison is mainly interpreted before and inside the shallow region, where the crest is still clearly identified. The abrupt loss of the numerical crest after the shallow region is not interpreted as a physical disappearance of the wave, but as a limitation of the crest-detection procedure used in the post-processing.
Figures 9, 10 and 11 show the amplitude profiles obtained for the three equations. The WKB curves are included as a reference. The agreement is not expected to be exact, because the numerical crest tracking is affected by the finite width of the wave packet and by the strong velocity variation near the reef. Nevertheless, the sign and relative strength of the amplitude variation are correctly reproduced.

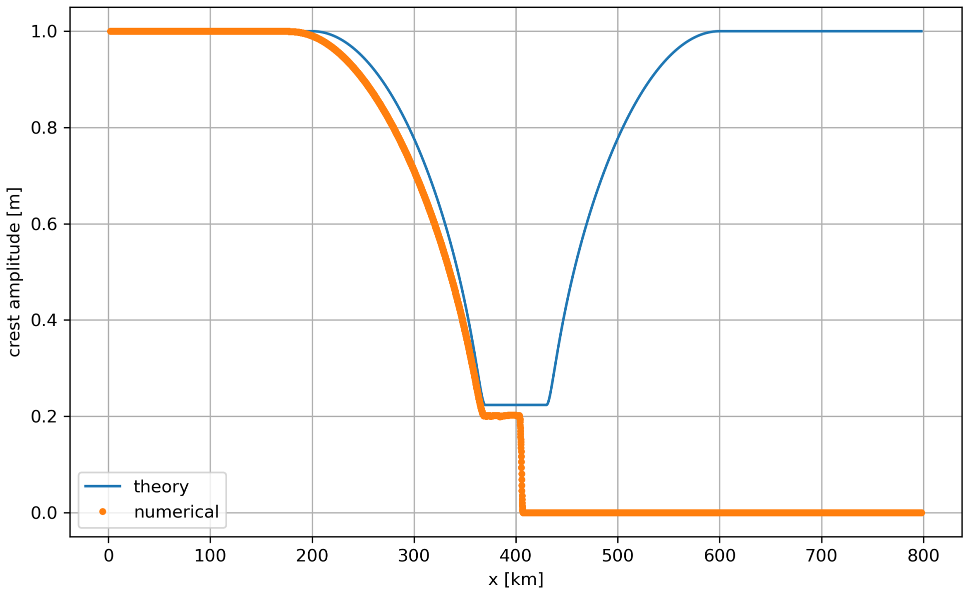


Figure 9: Comparison between the numerical first-crest amplitude and the WKB prediction for equation A. The amplitude decreases in the shallow region, consistently with the scaling $a_A(x) \propto h(x)^{1/4}$.

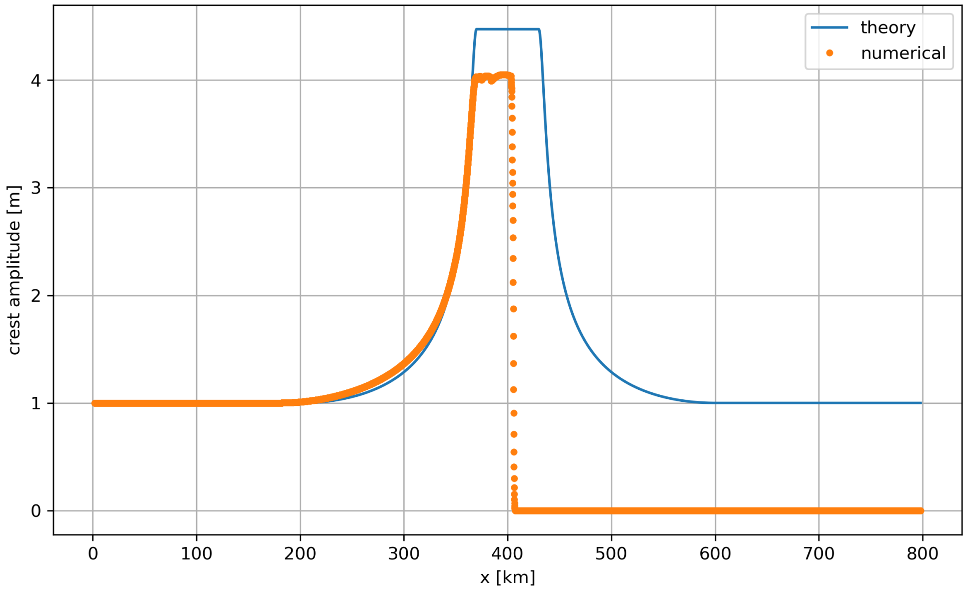


Figure 10: Comparison between the numerical first-crest amplitude and the WKB prediction for equation B. The amplitude increases moderately in the shallow region, consistently with the shoaling scaling $a_B(x) \propto h(x)^{-1/4}$.

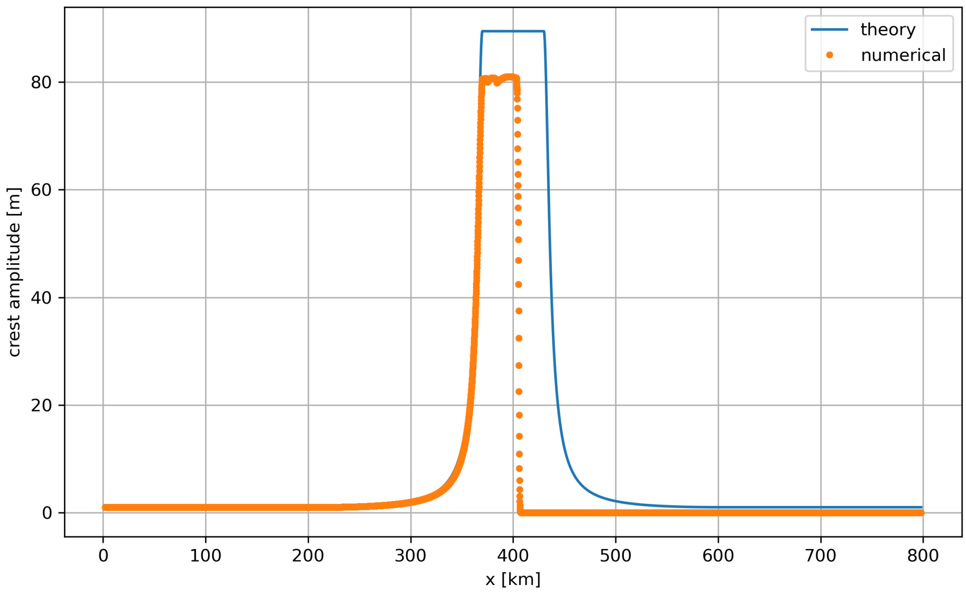


Figure 11: Comparison between the numerical first-crest amplitude and the WKB prediction for equation C. The amplification is much stronger than for equation B, consistently with the scaling $a_C(x) \propto h(x)^{-3/4}$.

The results show that the three equations do not describe the same amplitude evolution in an inhomogeneous medium. Equation A gives a decrease of amplitude in the shallow region, whereas equations B and C give an amplification. The amplification obtained with equation C is much larger than with equation B, in agreement with the stronger WKB scaling $h^{-3/4}$. Therefore, even if the three equations coincide for constant velocity, their behaviour in a variable medium is significantly different.

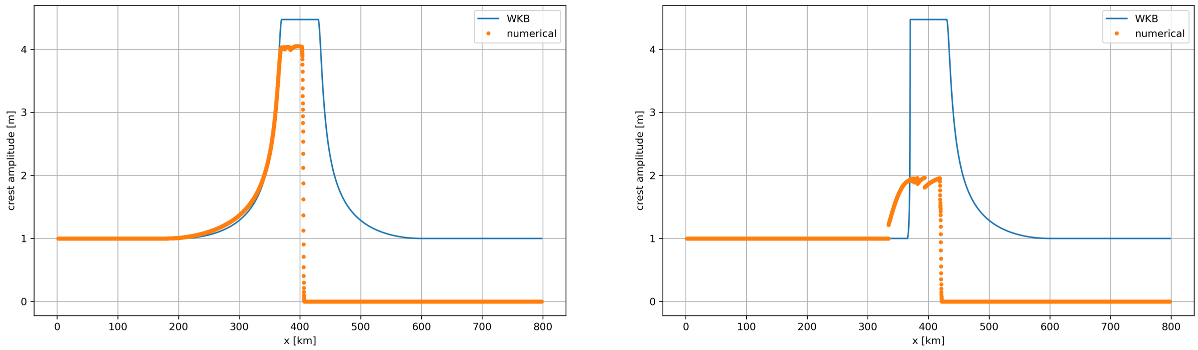
Among the three models, equation B is the most physically relevant for the tsunami problem, because it gives the expected shoaling behaviour: as the wave enters shallower water and its speed decreases, its amplitude increases.

3.2.2 b) Influence of the steepness of the reef

We now focus on equation B and study the effect of changing the steepness of the bathymetry. The parameter x_a is varied while x_b is kept fixed. Therefore, the quantity

$$x_b - x_a$$

measures the width of the transition from the deep-water region to the shallow-water region. A large value of $x_b - x_a$ corresponds to a smooth transition, whereas a small value corresponds to a more abrupt reef.
This test is useful because the WKB approximation assumes that the medium varies slowly compared with the local wavelength. Therefore, we expect the WKB prediction to be more accurate when the bathymetry varies smoothly, and less accurate when the transition is abrupt. Figure 12 compares two representative cases. In the first one, the transition is wide and the numerical amplitude follows the WKB prediction more closely. In the second one, the transition is much sharper and the numerical result remains further below the WKB plateau. This does not contradict the WKB prediction; it shows that the assumptions required for WKB are not fully satisfied when the bathymetry changes abruptly.



(a) Smooth transition

(b) Abrupt transition

Figure 12: First-crest amplitude for equation B for two different bathymetry transitions. For the smooth transition, the numerical amplitude approaches the WKB prediction more closely. For the abrupt transition, the numerical amplitude is lower and the WKB approximation is less accurate.

To quantify this effect, we measured the amplitude of the first crest on the reef for several values of $x_b - x_a$. The result is shown in Figure 13. The dashed horizontal line corresponds to the WKB prediction for the reef amplitude,

$$a_{\mathrm{WKB}} = A \left(\frac{h_R}{h_L} \right)^{-1/4} . \tag{44}$$

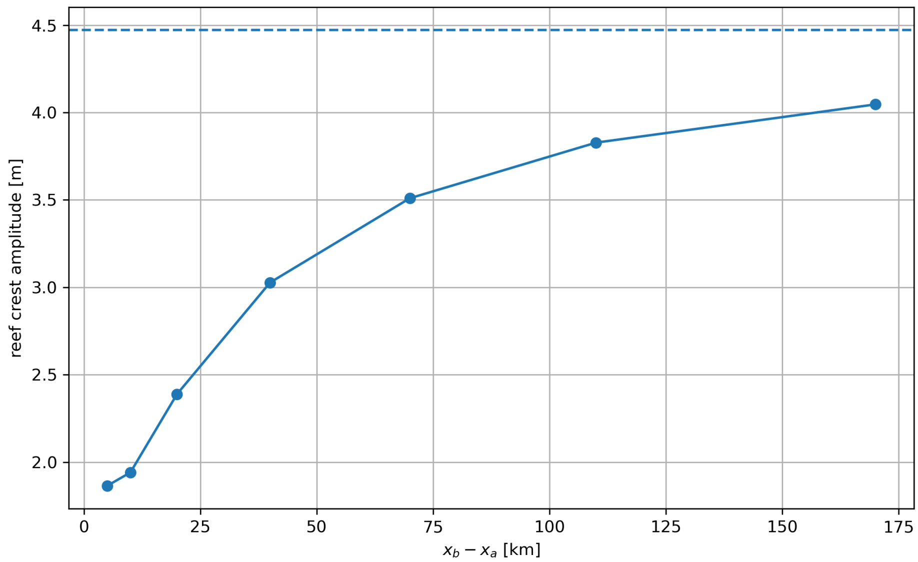


Figure 13: Numerical reef amplitude as a function of the transition width $x_b - x_a$ for equation B. The dashed line corresponds to the WKB prediction. As the transition becomes smoother, the numerical amplitude increases and approaches the WKB value from below.

The numerical results show a clear trend. For small values of $x_b - x_a$, the reef is abrupt and the measured amplitude is significantly below the WKB prediction. As $x_b - x_a$ increases, the bathymetry varies more gradually and the numerical amplitude increases toward the WKB value. This confirms that the WKB approximation is valid mainly when the propagation medium varies slowly.

This behaviour is physically consistent. When the depth changes smoothly, the wave adapts progressively to the local propagation speed, and the amplitude follows the local WKB shoaling law. When the depth changes abruptly, part of the wave is scattered or reflected, and the local WKB amplitude law no longer gives an accurate quantitative prediction.

3.3 Supplementary material

We now derive the WKB predictions for the three equations in the case of a slowly varying propagation speed $u(x)$. We consider a high-frequency ansatz of the form

$$f(x,t) = a(x) \exp\left[i\omega\left(t - \tau(x)\right)\right], \tag{45}$$

where $a(x)$ is a slowly varying amplitude and $\tau(x)$ is the travel-time function. The local wavenumber is

$$k(x) = \omega \tau'(x).$$

At leading order in ω, the three equations give the same eikonal equation,

$$u^2(x) \left(\tau'(x)\right)^2 = 1.$$

Therefore,

$$\tau'(x) = \frac{1}{u(x)}, \qquad k(x) = \frac{\omega}{u(x)}. \tag{46}$$

This shows that, in the WKB approximation, the local phase speed is indeed $u(x)$. Since in the shallow-water approximation

$$u(x) = \sqrt{gh(x)},$$

the wave speed is expected to follow the local water depth profile.

The next order in the WKB expansion gives the transport equation for the amplitude. This transport equation depends on the form of the wave equation.
For equation A,

$$\frac{\partial^2 f}{\partial t^2} = u^2 \frac{\partial^2 f}{\partial x^2},$$

the order-ω terms give

$$2a'(x)\tau'(x) + a(x)\tau''(x) = 0.$$

This can be written as

$$\frac{d}{dx}\left(a^2\tau'\right) = 0.$$

Using $\tau' = 1/u$, we obtain

$$a_A(x) \propto \sqrt{u(x)}.$$

Since $u(x) = \sqrt{gh(x)}$, this gives

$$a_A(x) \propto h(x)^{1/4}. \tag{47}$$

For equation B,

$$\frac{\partial^2 f}{\partial t^2} = \frac{\partial}{\partial x}\left(u^2 \frac{\partial f}{\partial x}\right),$$

the WKB transport equation becomes

$$u^2\left(2a'\tau' + a\tau''\right) + \left(u^2\right)' a\tau' = 0.$$

Using $\tau' = 1/u$, this simplifies to

$$2ua' + au' = 0.$$

Hence,

$$\frac{a'}{a} = -\frac{1}{2}\frac{u'}{u},$$

and therefore

$$a_B(x) \propto u(x)^{-1/2}.$$

Since $u(x) = \sqrt{gh(x)}$, we obtain

$$a_B(x) \propto h(x)^{-1/4}. \tag{48}$$

For equation C,

$$\frac{\partial^2 f}{\partial t^2} = \frac{\partial^2}{\partial x^2}\left(u^2 f\right),$$

we apply the WKB expansion to the product $u^2 f$. The transport equation can be written as

$$\frac{d}{dx}\left[\left(u^2 a\right)^2 \tau'\right] = 0.$$

Using again $\tau' = 1/u$, we get

$$u^2 a \propto \sqrt{u},$$

and therefore

$$a_C(x) \propto u(x)^{-3/2}.$$

Since $u(x) = \sqrt{gh(x)}$, this gives

$$a_C(x) \propto h(x)^{-3/4}. \tag{49}$$

The WKB predictions for the three equations can therefore be summarized as

$$\boxed{a_A(x) \propto h(x)^{1/4}, \qquad a_B(x) \propto h(x)^{-1/4}, \qquad a_C(x) \propto h(x)^{-3/4}.} \tag{50}$$

These results explain why the same bathymetry profile can produce very different amplitude variations depending on the equation used. Equation A predicts an increase of the amplitude in deeper water, while equations B and C predict an increase of the amplitude in shallower water. The effect is strongest for equation C, for which the amplitude scales as $h^{-3/4}$.

For equation B, the implementation used in the main part of the report discretizes the expanded form

$$\frac{\partial}{\partial x}\left(u^2\frac{\partial f}{\partial x}\right) = u^2\frac{\partial^2 f}{\partial x^2} + \left(u^2\right)'\frac{\partial f}{\partial x}.$$

An alternative is to discretize the conservative form directly. Defining

$$q(x) = u^2(x),$$

we can write

$$\frac{\partial^2 f}{\partial t^2} = \frac{\partial}{\partial x}\left(q\frac{\partial f}{\partial x}\right).$$

A conservative finite-difference discretization is then

$$\left[\frac{\partial}{\partial x}\left(q\frac{\partial f}{\partial x}\right)\right]_i \simeq \frac{1}{\Delta x}\left[q_{i+1/2}\frac{f_{i+1}-f_i}{\Delta x} - q_{i-1/2}\frac{f_i - f_{i-1}}{\Delta x}\right], \tag{51}$$

where the interface values are obtained by arithmetic interpolation:

$$q_{i+1/2} = \frac{q_i + q_{i+1}}{2}.$$

This method discretizes the flux qf_x as a whole, instead of first applying the product rule and then discretizing the two resulting terms separately. In smooth regions, both approaches are expected to be second-order accurate. However, when $u(x)$ varies rapidly, the conservative form is expected to be more robust because it better represents the continuity of the flux across neighbouring cells.

The update rule associated with the conservative discretization is

$$f_i^{n+1} = 2f_i^n - f_i^{n-1} + \Delta t^2\frac{1}{\Delta x}\left[q_{i+1/2}\frac{f_{i+1}^n - f_i^n}{\Delta x} - q_{i-1/2}\frac{f_i^n - f_{i-1}^n}{\Delta x}\right]. \tag{52}$$

To compare both approaches, we ran the tsunami configuration using equation B with the expanded discretization and with the conservative discretization. The same numerical parameters were used in both cases:

$$n_x = 1600, \qquad \beta_{\mathrm{CFL}} = 0.9, \qquad t_{\mathrm{fin}} = 5000\,\mathrm{s}.$$

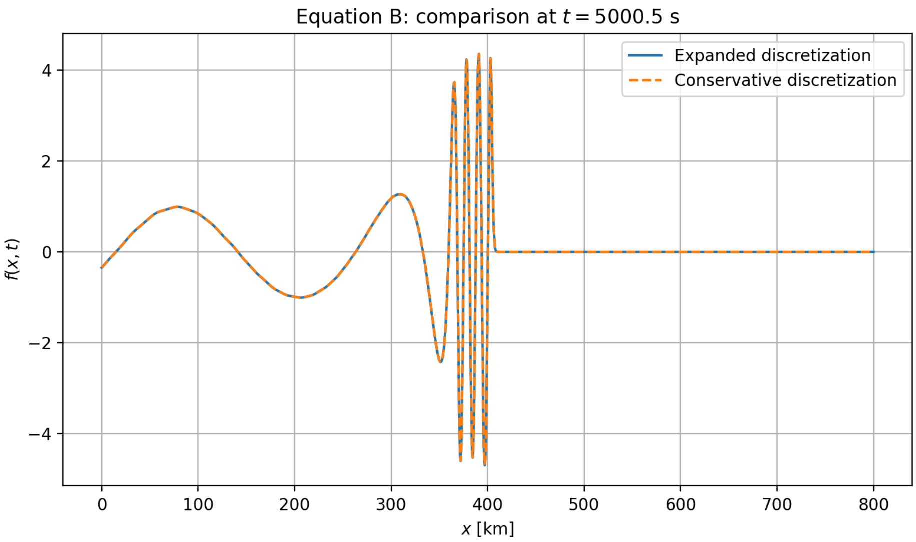


Figure 14: Comparison between the expanded and conservative discretizations of equation B at $t = 5000.49\,\mathrm{s}$. The two profiles are almost identical in the deep-water regions. Small differences appear mainly near the region where the bathymetry varies rapidly and the wave interacts with the reef.

We quantify the difference between the two numerical solutions using

$$\delta(t) = \left[\int_0^L \left(f_{\text{expanded}}(x,t) - f_{\text{conservative}}(x,t) \right)^2 dx \right]^{1/2}. \tag{53}$$

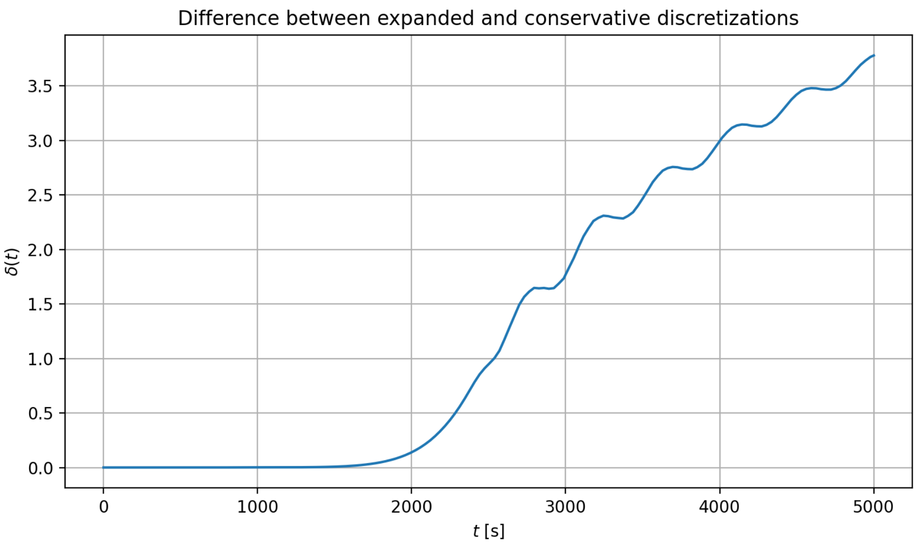


Figure 15: Time evolution of the difference $\delta(t)$ between the expanded and conservative discretizations of equation B. The difference remains small before the wave reaches the region of variable bathymetry and then increases as the wave interacts with the reef.

The maximum difference over the simulation was

$$\max_t \delta(t) = 3.7806. \tag{54}$$

At the final time, we obtained

$$\delta(t_{\text{fin}}) = 3.7806. \tag{55}$$

The maximum pointwise difference at the final time was

$$\|f_{\text{expanded}} - f_{\text{conservative}}\|_\infty = 2.8440 \cdot 10^{-2}. \tag{56}$$

The fact that the maximum of $\delta(t)$ is reached at the final time indicates that the difference between the two discretizations accumulates as the wave interacts with the variable bathymetry. The comparison shows that the two discretizations give similar results in smooth regions, but differences appear near the parts of the domain where the velocity changes rapidly. This is expected because the conservative discretization treats the flux qf_x directly, while the expanded discretization separately approximates qf_{xx} and $q_x f_x$. Therefore, the conservative form is generally preferable when the medium is strongly inhomogeneous or when the grid resolution is limited.

4 Conclusions

In this exercise, we implemented explicit finite-difference schemes for three forms of the one-dimensional wave equation. In the constant-speed case, the three equations reduce to the same standard wave equation, and the numerical simulations reproduced the expected behaviour: fixed boundaries reflect waves with a sign inversion, free boundaries reflect them without sign inversion, and outgoing-wave boundary conditions strongly reduce artificial reflections.
The eigenmode analysis showed that the natural frequencies of the fixed–fixed system are given by

$$\omega_n = \frac{n\pi u}{L}.$$

The numerical frequency scan confirmed this result, since the largest energy responses appeared close to the analytical eigenfrequencies. The stability study also verified the expected CFL condition. For $\beta_{\text{CFL}} \leq 1$, the numerical solution remained bounded, whereas for $\beta_{\text{CFL}} > 1$, the solution became unstable.
In the tsunami application, the propagation speed was no longer constant, but depended on the bathymetry through $u(x) = \sqrt{gh(x)}$. In this case, equations A, B and C no longer gave the same result. The WKB analysis predicted different amplitude scalings for the three models, and the simulations confirmed that the amplitude evolution depends strongly on the chosen equation. Equation B gave the most physically relevant behaviour for the tsunami problem, with an increase of the wave amplitude in shallow water.
Finally, the study of the reef steepness showed that the WKB approximation becomes more accurate when the bathymetry varies smoothly. When the transition to shallow water is abrupt, the numerical amplitude differs more from the WKB prediction, which is consistent with the fact that WKB theory assumes a slowly varying medium.

5 References

1. Exercise statement: *Physique Numérique – Exercice 5, Equation d'onde dans un milieu inhomogène : modes propres et propagation d'une vague de tsunami.*

2. L. Villard, *Physique Numérique.*

3. Use of LLM tools (ChatGPT) for language correction and report rephrasing.